Final Project Report

Online Smart Parking Management System

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1. Introduction and Project Significance

Main Problem: Finding parking spaces in densely populated cities leads to drivers wasting time, a significant increase in traffic congestion, elevated fuel consumption, and environmental pollution.

Project Environment: This system is designed for public parking lots, large urban multi-story parking structures, and curbside parking spaces.

Necessity of the Internet of Things (IoT): Unlike traditional systems where a driver only realizes a parking lot is full upon physical arrival, an IoT-based system enables real-time monitoring of each parking spot and web-based online reservation. This eliminates vehicle wandering and completely optimizes space management.

2. System Requirements

- **Functional Requirements:** User registration, wallet recharging, real-time viewing of parking space status, firm reservation of an empty spot for a specific time slot, reservation cancellation before the deadline, and an admin monitoring panel (manual sensor status overrides, reservation management, viewing reservation reports for various customers, etc.).
- **Qualitative Requirements:** High security in storing identity and financial data (hashing encryption), high responsiveness in status updates (optimized polling), and database stability under concurrent loads.
- **Network Connectivity of Things:** Sensor nodes installed at each parking spot connect to the network. Status data regarding vehicle presence (0 for empty and 1 for occupied) is collected and sent to the central server via an API.

3. Required Equipment and Technologies

3.1. Hardware Component

- **Sensors:** HC-SR04 ultrasonic modules or infrared (IR) sensors for continuous distance measurement and physical detection of vehicle presence or absence.

- **Actuators:** Mini servo motor modules to automatically open and close gates or barriers at the parking entrance/exit.

- **Central Processing Unit:** Microcontrollers such as ESP32 or ESP8266 nodes. These boards are selected due to their built-in Wi-Fi module, low power consumption, optimized dimensions, and cost-effectiveness for edge nodes.

3.2. Software Component

- **Programming Language and Backend:** Python and the Flask framework.

- **Database:** SQLite smart relational database.

- **Frontend and Dashboard:** HTML, CSS, raw JavaScript along with the powerful Three.js library for 3D graphical rendering and display of the parking lot.

4. Overall System Architecture

The overall system architecture is based on a Client-Server and Edge-Server Internet of Things (IoT) structure. The main components include:

1. **Physical/Edge Layer:** Sensors collect and pre-process vehicle presence data.

2. **Network and Communication Layer:** Data transfer by sending web requests.

3. **Server Layer (Core API):** Processing business logic, preventing reservation conflicts, and validating the financial system.

4. **Storage Layer:** SQLite database for record persistence.

5. **Frontend/User Dashboard Layer:** Live 3D display for drivers and administrators.

5. Protocols and Communication Layer

- **Short-Range/Local Communication Protocol:** Sensor nodes deployed at each spot connect to the central parking modem or gateway using built-in Wi-Fi technology.

- **Data Transfer Protocol:** Data exchange between the central server and clients (and node simulators) is carried out over the standard HTTP protocol in the modern JSON format. The chosen REST API structure provides high exchange speed, good security, and full compatibility with network bandwidth and frontend processing.

6. Data Management, Processing, and Cloud

- **Data Processing Location:** Processing is performed using a hybrid approach. Initial detection of vehicle presence occurs at the edge level (by ESP boards), and compressed status data is sent to the server. More complex processing, such as overlap checks for timing and wallet balance deduction, takes place on the server side.

- **Dashboard and Data Visualization:** The system utilizes a smart web-based Single Page Application (SPA) dashboard. Users can easily view a live, animated 3D map of the parking lot. Car movement animations toward designated spots are implemented using a client simulator framework and Three.js.

7. Target Users and Customers

1. Urban vehicle drivers who need to find an empty parking space quickly and hassle-free.
2. Owners and management companies of large public and private parking lots for online traffic and revenue monitoring.
3. Municipalities and urban traffic transport organizations for macro-level city traffic planning.

8. Value Proposition

This system offers the following competitive advantages over traditional methods:

- Ability to reserve a specific parking spot before departing for the destination.
- Fully visual, interactive, and 3D display of the parking lot map, creating a highly tangible user experience.
- Implementation of smart physical conflict resolution algorithms (releasing spaces occupied by unauthorized vehicles before the official reservation time begins).

9. Competitive Analysis and Innovation

- **Traditional SMS-Based Reservation Systems:** Simple functionality but lack live spot monitoring, leading to high error and conflict rates.
- **Curbside Digital Parking Meters:** Only record presence but lack remote reservation and centralized 3D monitoring capabilities.
- **Smart Systems Based on Image Processing:** High camera and processing costs, prone to disruptions in absolute darkness.
- **Our Project's Distinction:** Integration of a comprehensive internal financial system with a native sensor simulator, real-time 3D live graphical display, and significantly lower hardware implementation costs compared to image processing alternatives.

10. Implementation Feasibility

Operational Evaluation: All backend code, database logic, and frontend 3D dashboards have been successfully implemented. Given the abundance of low-cost components (ESP32 and ultrasonic sensors) in the local market, physical implementation of the design within an academic semester is entirely achievable and executable.

Alternative Scenario: A native and dedicated simulation module has been designed and implemented in this project inside the `sensor_sim.py` file. This script precisely simulates the behavior of IoT nodes, generating network traffic, random car entry/exit data, and managing sensor states to ensure the platform operates without the slightest interruption in the absence of physical hardware.

11. Challenges, Limitations, and Power Management

- **Challenge 1:** Potential loss of Wi-Fi connectivity in the concrete environments of underground multi-story parking structures.
  Solution: Implementing a temporary caching system at the edge (ESP).
- **Challenge 2:** Unauthorized occupation of a reserved spot by other drivers.
  Solution: Implementing software-driven release logic and sending an alert signal or automatically changing the sensor status 30 seconds before the official reservation starts in the system code.

12. Initial Cost Estimation

The estimated budget for constructing the prototype is presented in the table below:

Equipment and Service Description	Estimated Cost (Tomans)
Central Processing Board (ESP32 NodeMCU)	500,000
Ultrasonic and Infrared Sensors (5 units)	200,000
Actuators and Enclosures (Gate Servo Motor, Wires, Breadboard, Box)	400,000
Communication and Server Hosting Costs	500,000
Total Estimated Budget	**1,600,000 to 2,000,000**